2 Determination of mass of D-glucose in a sample

The Bertrand method is a classical redox titration technique used to determine the amount of reducing sugars, such as D-glucose.

In this method, a sample of D-glucose, $C_6H_{12}O_6$, is reacted with excess alkaline copper(II) solution, Cu^{2+}, to form insoluble copper(I) oxide, Cu_2O.

equation 1 $C_6H_{12}O_6 + 2Cu^{2+} + 5OH^- \rightarrow C_6H_{11}O_7^- + Cu_2O + 3H_2O$

The Cu_2O is filtered and dissolved in 100 cm^3 acidic iron(III) solution, Fe^{3+}.

equation 2 $Cu_2O + 2Fe^{3+} + 2H^+ \rightarrow 2Cu^{2+} + 2Fe^{2+} + H_2O$

The resulting solution is then diluted to 1 dm^3. The diluted solution is labelled **FA 3**.

The amount of Fe^{2+} present in **FA 3** is then determined by titration with potassium manganate(VII), $KMnO_4$, in acidic medium. The end-point of the titration occurs when the presence of unreacted MnO_4^- causes the colour of the solution to become pale pink.

equation 3 $5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$

You are to determine the amount of Fe^{2+} present in **FA 3** and hence determine the mass of D-glucose in the sample.

FA 1 is 0.0200 mol dm^{-3} potassium manganate(VII), $KMnO_4$.
FA 2 is 1.0 mol dm^{-3} sulfuric acid solution, H_2SO_4.

(a) (i) Procedure

1. Fill the burette with **FA 1**.
2. Pipette 25.0 cm^3 of **FA 3** into a clean 250 cm^3 conical flask.
3. Use a measuring cylinder to add 10.0 cm^3 of **FA 2** to the conical flask.
4. Run **FA 1** from the burette into the conical flask. The end-point is reached when the **first permanent** trace of pale pink colour is seen.
5. Record your titration results, to an appropriate level of precision, in Table 2.1.
6. Repeat points 2 to 5 until consistent titre values are obtained.

Table 2.1

final burette reading / cm^3					
initial burette reading / cm^3					
volume of **FA 1** used / cm^3					

[2]

(ii) From your titration results, obtain a suitable volume of **FA 1** to be used in your calculations. Show clearly how you obtained this volume.

volume of **FA 1** = ... cm^3 [3]

(b) **(i)** Calculate the amount of Fe^{2+} in 1 dm^3 of **FA 3**.

amount of Fe^{2+} in 1 dm^3 of **FA 3** = ...
[3]

(ii) Calculate the mass of D-glucose, $C_6H_{12}O_6$, in the sample.
[A_r: H, 1.0; C, 12.0; O, 16.0]

mass of glucose in sample = ... [2]

(c) A student repeated the procedure in **(a)** and obtained an average titre value of 19.50 cm^3.

The errors (uncertainties) associated with **each reading** using pipette and burette are ±0.10 cm^3 and ±0.05 cm^3 respectively.

Calculate the maximum total percentage error (uncertainty) of this average titre volume.

[1]

[Total: 11]